MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2024-25) -- SEMESTER 5

COURSE: DATABASE MANAGEMENT SYSTEMS (DBMS) (PCC-CS501)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: PCC-CS501

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Differentiate between file processing system and DBMS.

Q1. (b) Define three-tier ANSI-SPARC database architecture and physical/logical data independence.

Q1. (c) What is candidate key, primary key, super key, and foreign key?

Q1. (d) Define functional dependency and Armstrong inference axioms.

Q1. (e) What is lossless-join decomposition and dependency preservation?

Q1. (f) State ACID properties of database transactions with examples.

Q1. (g) What is dirty read problem and unrepeatable read problem?

Q1. (h) Explain Two-Phase Locking (2PL) protocol: Growing phase vs Shrinking phase.

Q1. (i) What is write-ahead logging (WAL) rule?

Q1. (j) Differentiate between Dense index and Sparse index in B+ tree.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Draw an ER diagram for a Hospital Management System with doctors, patients, wards, and treatments.

Q2. (b) Given relation R(A, B, C, D, E) with FDs: {A -> BC, CD -> E, B -> D, E -> A}. Find all candidate keys and highest normal form.

Q2. (c) Normalize relation R(A, B, C, D, E, F) with FDs: {AB -> C, C -> D, D -> E, E -> F} into 3NF and BCNF.

Q2. (d) Explain conflict serializability. Test whether given schedule S is conflict serializable by drawing precedence graph.

Q2. (e) Construct a B+ tree of order 4 by inserting keys: 5, 15, 25, 35, 45, 55, 40, 30, 20 with node splitting.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Write SQL queries for Company schema: nested subqueries, GROUP BY, HAVING, and correlated subqueries. [10 Mark

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(b) Explain Relational Algebra fundamental operations: Select, Project, Cartesian Product, Union, Set Difference, and Natural Join.

Q4. (a) Explain multi-valued dependencies and 4NF with an illustrative real-world relation. [10 Marks]

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(b) Explain 5NF (Project-Join Normal Form) and domain-key normal form (DKNF). [10 Marks]

Q5. (a) Explain Timestamp Ordering protocol and Thomas Write Rule for concurrency control. [10 Marks]

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(b) Explain deadlocks in DBMS: Wait-Die scheme vs Wound-Wait scheme with timestamps. [10 Marks]

Q6. (a) Explain ARIES database recovery algorithm (Analysis, Redo, and Undo passes) with log records. [10 Marks]

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(b) Explain checkpointing in recovery and immediate vs deferred database modification. [10 Marks]

Q7. (a) Compare B-Tree and B+ Tree indexing based on height, disk I/O, and range query efficiency. [10 Marks]

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(b) Explain query optimization stages: parsing, query tree heuristics, and cost-based evaluation. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****